

FINAL

CSE 420

COMPILER DESIGN

FALL 2023

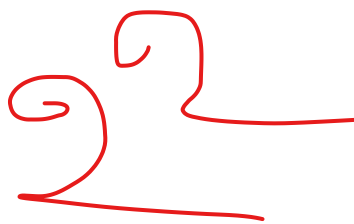
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Section: 07

Date: 18.12.2023

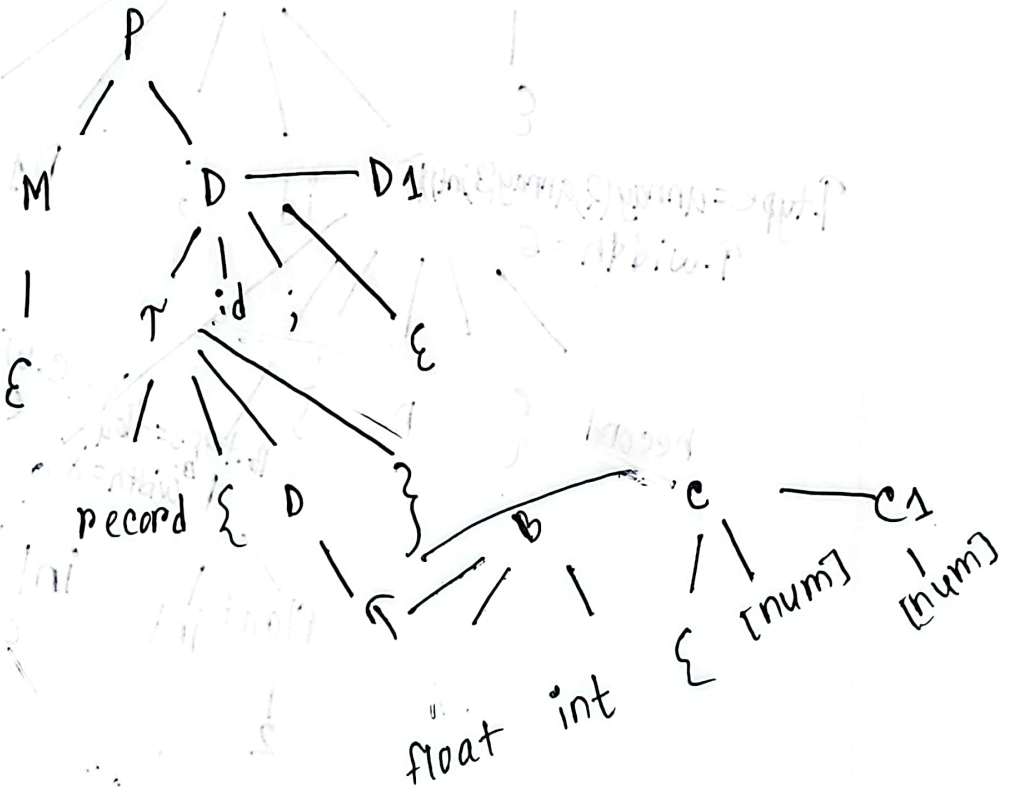
Signature: Sanzana

A red handwritten signature, possibly reading 'Sanzana', is written in a stylized, cursive manner.

Answer to Question No:-01

a.

Step-01: Drawing a parse tree



Given input = record { float [2][3] a; int b; } P;

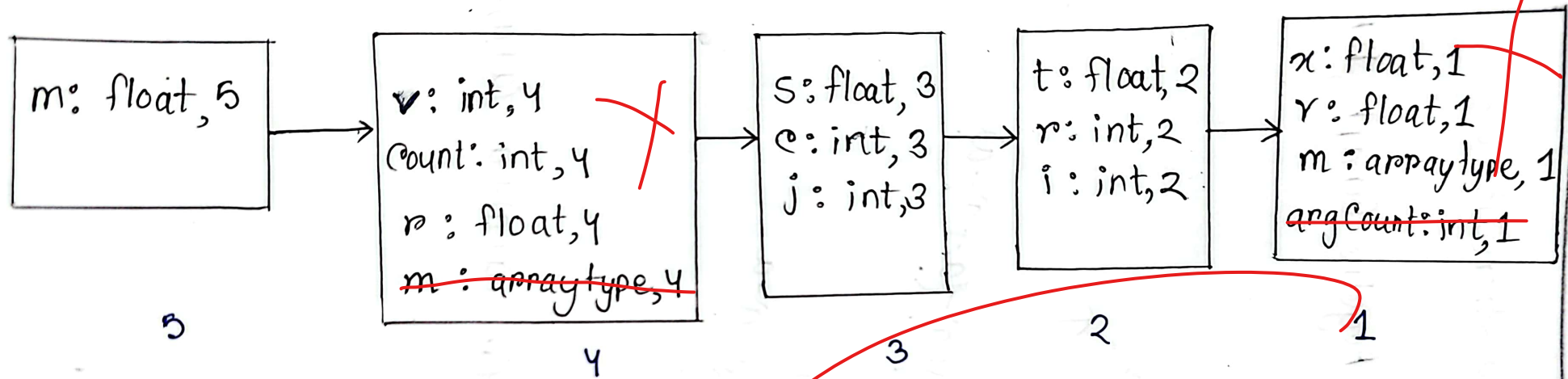
```
record { float [2][3] a; int b; } r;
```



Answer to Question No: 02

```
1 { float x = 0.0;
    float r = 0.0;
    int [10][20] m;
    void main (int argc, char *[] argv) {
        x = toFloat(argv[1]);
        m = readMatrixFromFile(argv[2]);
        float t = 1.0;
2 { int p = 10;
    for (int i = 0; i < c; i += 1) {
        float s = 0.0;
        int c = 20;
3 { for (int j = 0; j < c; j += 1) {
        int v = m[i][j];
        int count = 0;
        float p = 1.0;
4 { while (count < r) {
            count += 1;
5 { float m = sqrt(x);
        p = p * m;
```

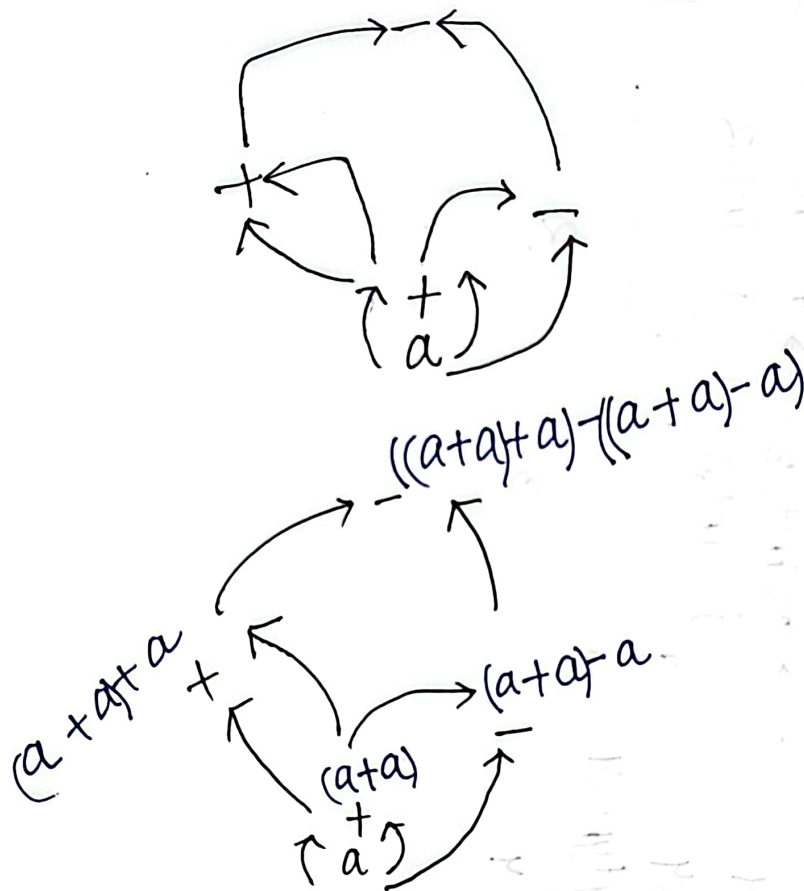
Single list of hash-tables:



Answer to Question No: 03

a.

Given Directed Acyclic Graph, DAG,



Three address code: for $(a+a)+a-(a+a)-a$

$$t_1 = a+a$$

$$t_2 = t_1 + a$$

$$t_3 = t_1 - a$$

$$t_4 = t_2 - t_3$$

(Ans:-)

b. Quadruples

	OP	arg1	arg2	result
(0)	+	a	a	t ₁
(1)	+	t ₁	a	t ₂
(2)	-	t ₁	a	t ₃
(3)	-	t ₂	t ₃	t ₄

 Triples

	OP	arg1	arg2
(0)	+	a	a
(1)	+	(0)	a
(2)	-	(0)	a
(3)	-	(1)	(2)